GD

$$x_{t+1} = x_t - \alpha \nabla f(x_t)$$

+Momentum

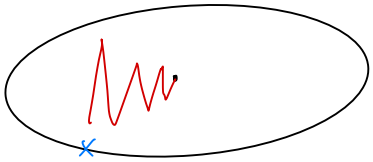
$$v_{t+1} = \rho v_t - \alpha \nabla f(x_t)$$

$$x_{t+1} = x_t + v_{t+1}$$

Nesterov Momentum

$$v_{t+1} = \rho v_t - \alpha \nabla f(x_t + \rho v_t)$$

$$x_{t+1} = x_t + v_{t+1}$$



- oscillation ⇓ (진동 ⇓)
- fast convergence (수렴 ⇑)
- escape local min (탈출)

불필요한 왕복운동

smooth.

⇒ 방향 컨트롤.

AdaGrad

$$x = x - \alpha \cdot \frac{\nabla f(x_t)}{\sqrt{\sum (\nabla f(x_i))^2} + \varepsilon}$$

$\| \nabla f(x_t) \| \Uparrow$ "가파르다"

- 가파르면 → 느리게
- 완만하면 → 빠르게

RMSProp

$$E[g^2]_t = \beta E[g^2]_{t-1} + (1-\beta) g_t^2 .$$

$$x = x - \frac{\eta g_t}{\sqrt{E[g^2]_t} + \varepsilon}$$

완만 → 빠르게!

경도를 → 늦게!

⚠ $\sqrt{\sum (\nabla f(x_i))^2} \to \infty$

⇒ 속도 컨트롤.

Adam

$$\boxed{x_{t+1} = x_t - \alpha \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon}}$$

(방향) $m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$

(속도) $v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2$

(편향 보정) $\hat{m}_t = \dfrac{m_t}{1-\beta_1^t}$

$m_0 = 0$

$v_0 = 0$

$\hat{v}_t = \dfrac{v_t}{1-\beta_2^t}$

pf. let $\nabla f = g$ (const)

$$m_k = (1-\beta_1^k) g \longrightarrow \hat{m}_k = g$$

$$v_k = (1-\beta_2^k) g^2 \longrightarrow \hat{v}_k = g^2 .$$